DATA SCIENCE & ARTIFICIAL INTELLIGENCE

& CS/IT





Calculus and Optimization

Maxima and Minima

Lecture No. 02



By- Dr. Puneet Sharma Sir

Recap of previous lecture



Topic

Maxima - Minima (Part 1)

Topics to be Covered



Topic MAXIMA-MINIMA (Part 2)

Maxima & Minima (Part-I) → UNIT

(Increasing & Decreasing func^n) → Chapter.

W.K.T that for $y = f(x)$, $\frac{dy}{dx} = f'(x) = \tan\theta$ = Slope of tangent at any Random Point x on $f(x)$

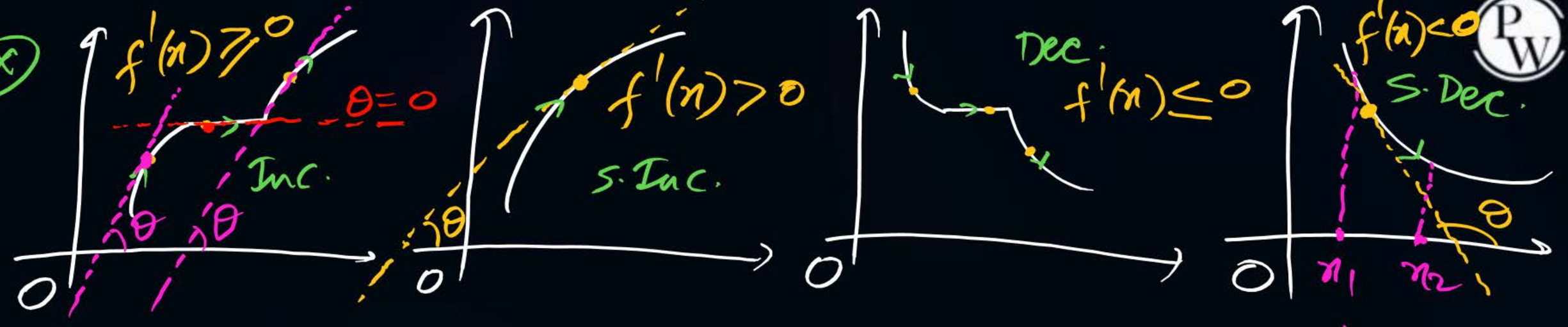


(1) for Acute angle tangent, $f'(x) > 0$

(2) " obtuse " " , $f'(x) < 0$

(3) for Horizontal tangent, $f'(x) = 0$

(4) for Vertical Tangent, $f'(x) = D.N.E$



eg $f(x_1) > f(x_2)\ \forall\ x_1 < x_2$; $x_1, x_2 \in D_f$ then $f(x)$ is S. Dec

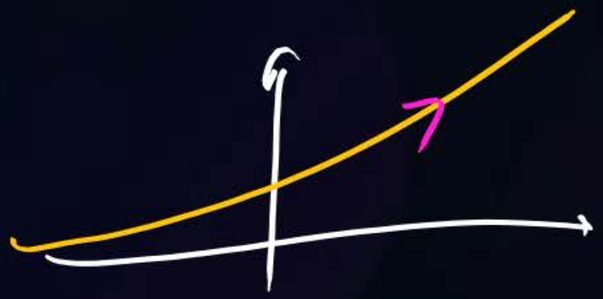
eg $f(x_1) \geq f(x_2)\ \forall\ x_1 < x_2$; $x_1, x_2 \in D_f$ then $f(x)$ is Decreasing

eg $f(x) = e^x$ is S. Inc

(M-II) $f(x) = e^x$, $D_f = (-\infty, \infty)$

$f'(x) = e^x$

$\because f'(x) > 0$ always so $f(x)$ is S. Inc.



Q: Find the Interval in which $f(x)=(x^2-4)^2$ Increases & Decreases?

Sol: $f(x)=(x^2-4)^2$; $(-\infty,\infty)$

$f'(x)=2(x^2-4)(2x)$

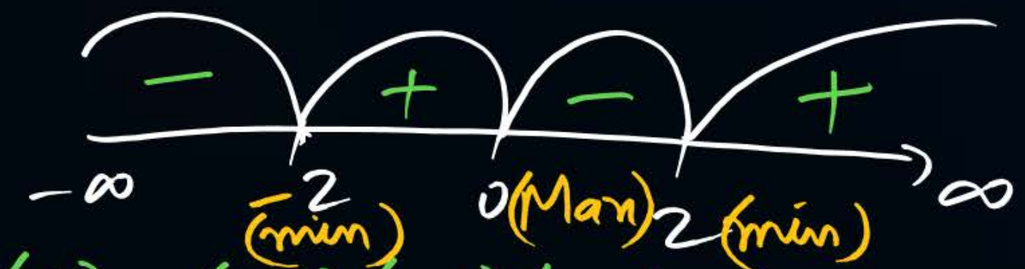
$f'(x)=4x(x-2)(x+2)$

T. Points of $f(x)$ are

$f'(x)=0$

$4x(x+2)(x-2)=0$

$x=-2,0,2$



$f'(-3)=(-ve)(-ve)(-ve)=-ve$

$f'(-1)=(-ve)(-ve)(+ve)=+ve$

$f'(1)=(+ve)(-ve)(+ve)=-ve$

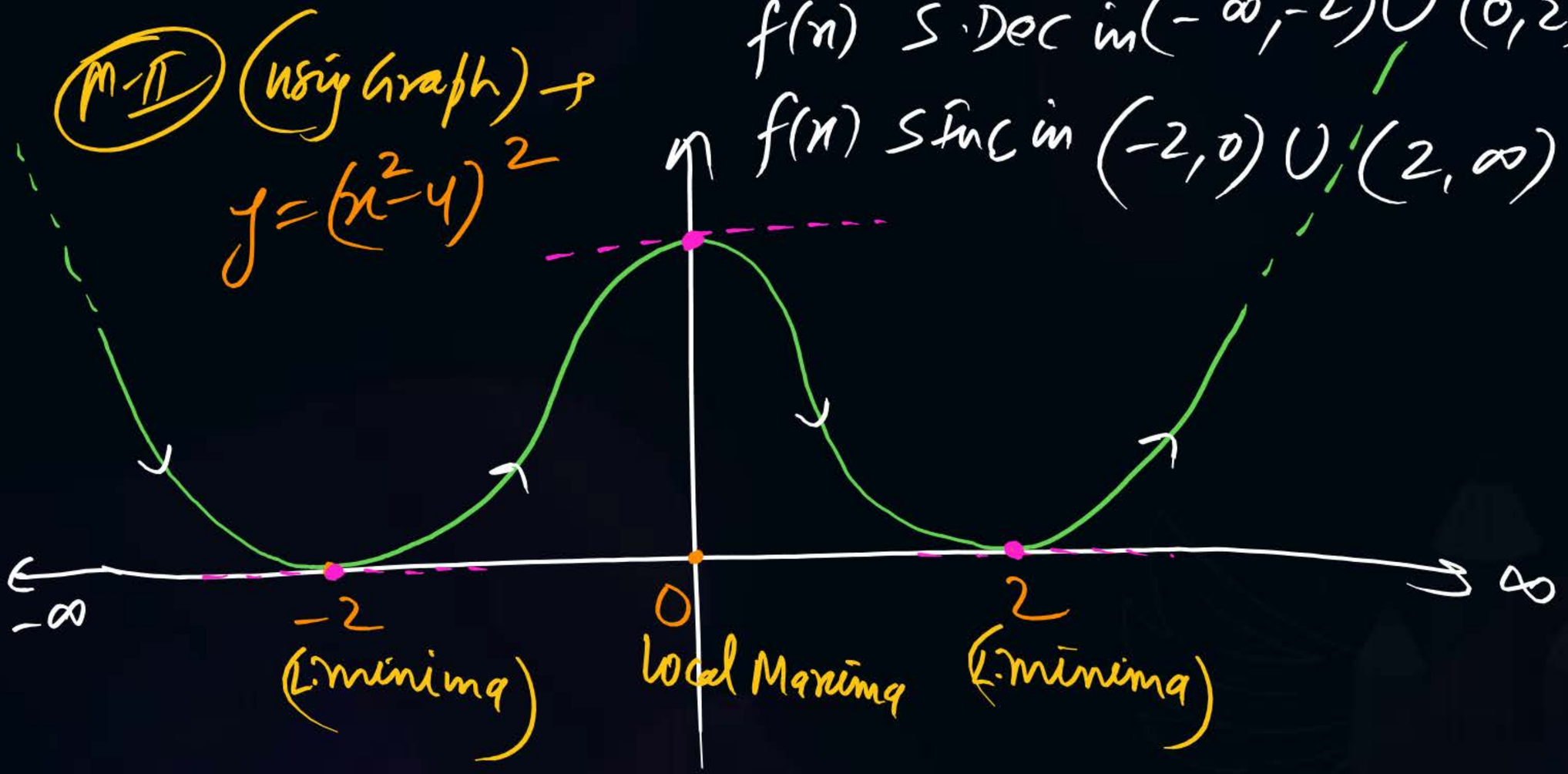
$f'(3)=(+ve)(+ve)(+ve)=+ve$

$f(x)$ S.Inc in $(-2,0)\cup(2,\infty)$

$f(x)$ SD in $(-\infty,-2)\cup(0,2)$

M-II (using Graph) →

$y = (x^2-4)^2$



(HW) Find the Interval of Increasing & Decreasing for $y = \frac{\log_e x}{x}$

Domain $= (0, \infty)$

Sol: $f(x) = \frac{\log_e x}{x}$

$$f'(x) = \frac{x\left(\frac{1}{x}\right) - \log_e x (1)}{(x)^2}$$

$$\boxed{f'(x) = \frac{1 - \log_e x}{x^2}}$$

$\because x > 0$ so For Turning Point.

$f'(x) = 0 \Rightarrow 1 - \log_e x = 0 \Rightarrow \boxed{x = e}$



$f'(1) = \frac{1 - \log_e 1}{1^2} = \frac{1-0}{1} = 1 = +ve$

$f'(e^2) = \frac{1 - \log e^2}{(e^2)^2} = \frac{1-2}{e^4} = -ve$

$f(x)$ is Inc in $(0, e)$

$f(x)$ is D in (e, ∞).

Analysis: ① $f'(x)=\frac{1-\ln x}{x^2}$, f

$f(x)$ is S.Inc in $(0,e)$

$f(x)$ is Inc in $(0,e]$ $\because f'(e)=0$

$f(x)$ is S.Dec in (e,∞)

$f(x)$ is Dec in $[e,\infty)$ $\because f'(e)=0$

② 

$y=f(x)=\frac{\ln x}{x}$, $D_f=(0,\infty)$

$f(1)=\frac{0}{1}=0$

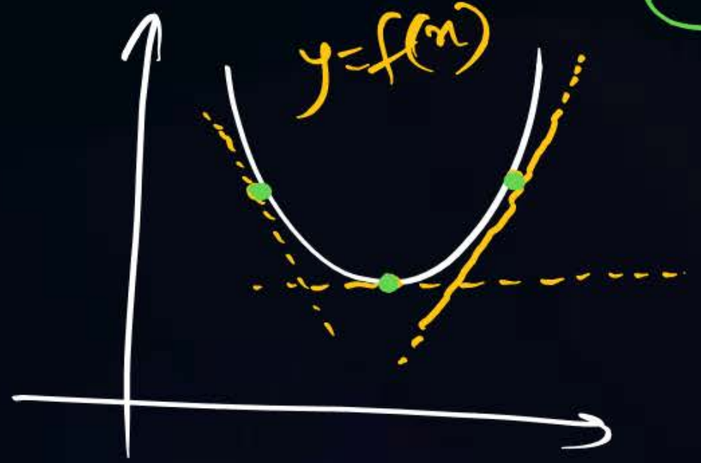
③ $f(x)=\ln_e x$, $(0,\infty)$



Information :

Concave upward Curve :

if Curve lies above the tangent always then Curve is called Concave upward Curve



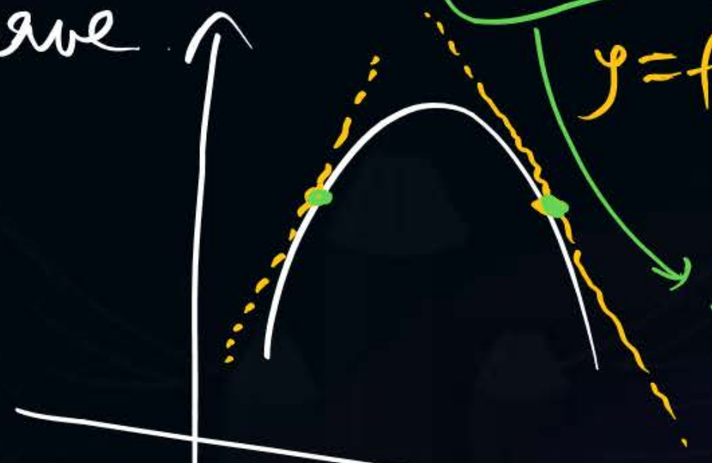
$\Downarrow$

$f''(x) > 0$

eg $f(x) = e^x$ is Concave upward Curve

Concave Downward Curve

if Curve lies below the tangent always then Curve is called Concave Downward Curve



eg $f(x) = \log_e x$ is Concave Downward Curve

Eg: $y = x^4$ is Concave upward.

LE

MFI $f(x) = x^4$, $f'(x) = 4x^3$, $f''(x) = 12x^2$

$\because f''(x) > 0 \Rightarrow f(x)$ is Concave upward.

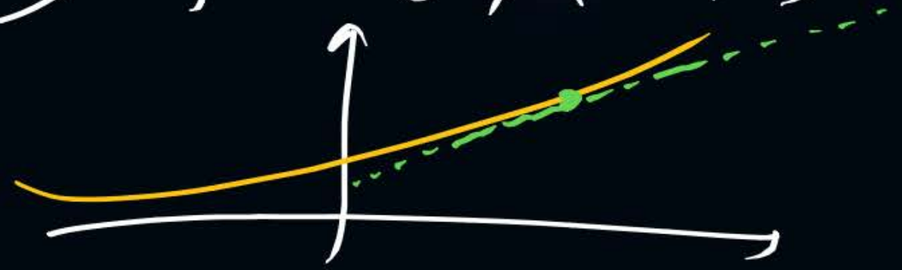
Justification of the Standard Result :-

$f(x) = x^4$ $\quad$ $f'(x) = 4x^3$ $\quad$ $f''(x) = 12x^2$



$f(x)$ is Concave up $\Rightarrow$ $f'(x)$ is S.Inc $\Rightarrow$ $f''(x) > 0 \Rightarrow f(x)$ is Concave upward.

(M-I) $f(x)=e^x$, $(-\infty,\infty)$



(M-II) $f(x)=e^x$

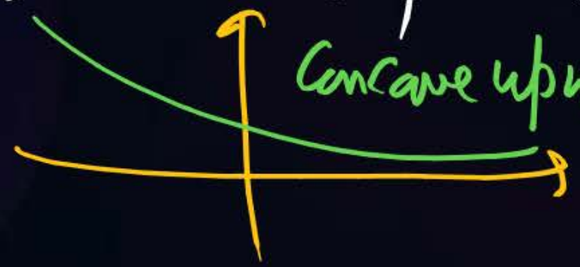
$f'(x)=e^x>0$

$\Rightarrow$ $f(x)$ is S.Inc.

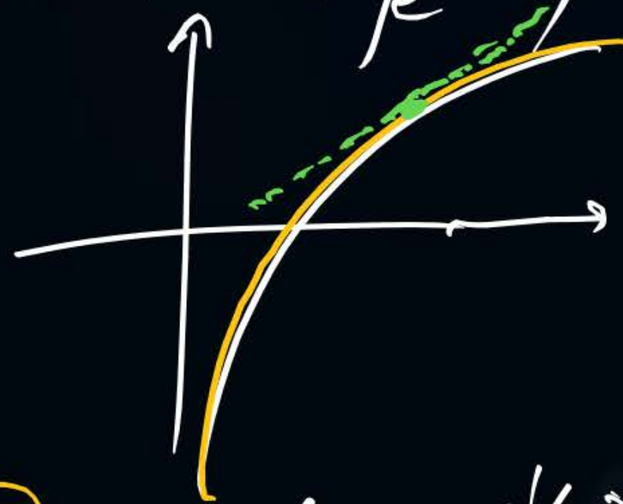
& $f''(x)=e^x>0$

$\Rightarrow$ $f(x)$ is Concave upward.

Ex $f(x)=e^{-x}$



(M-I) $f(x)=\log_e x$, $(0,\infty)$



(M-II) $f(x)=\log_e x$, $f'(x)=\frac{1}{x}$, $f''(x)=\frac{-1}{x^2}$

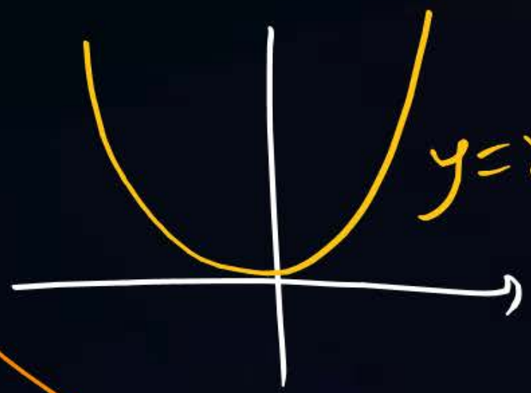
$\because f'(x)>0$ always $\Rightarrow f(x)$ is S.Inc.

$\because f''(x)<0$ " " " is Concave Down.

eg: $y = e^x, x^2, x^4, x^6$ ..... Concave upward Curve.

eg $y = \log x, -x^2, -x^4, -x^6$ .. Concave Downward Curve

4 $x^{1/2}, x^{1/4}, x^{1/6}$ ... " " " "

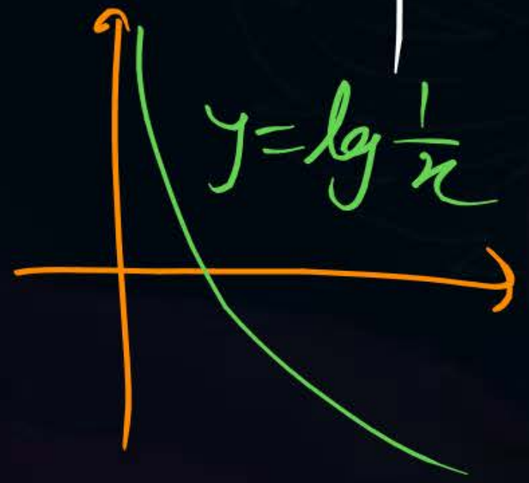






eg $y = \log\frac{1}{x} = \log x^{-1} = -\log x$

Concave upward.



Saddle point / Point of Inflexion →

"Those points where Curve changes it's Concavity are called Points of Inflexion"

Note: (1) those points where we are getting Neither Maxima, Nor minima are called Points of Inflexion (F)

(2) if ($x=\alpha$ is the point of Inflexion) $\Longrightarrow$ (At $x=\alpha$ we will get NOM.NM)

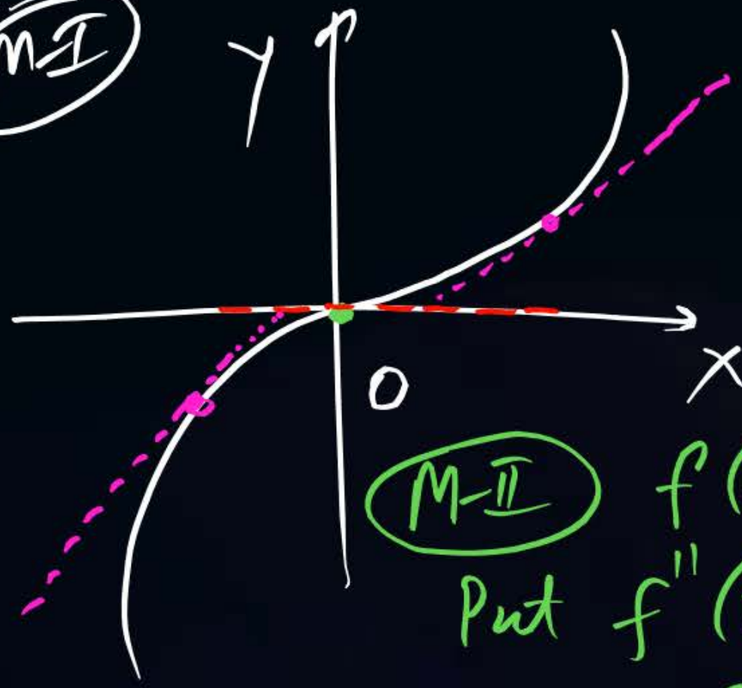
Shortcut: Put $f''(x)=0$ & Try to get x (Say it is $x=\alpha$)

if $f'''(\alpha)\neq 0 \Rightarrow x=\alpha$ is point of Inflexion

& if $f'''(\alpha)=0 \Rightarrow$ we can't say anything about α.

eg for the func^n $y=x^3$, $x=0$ is <u>Point of Inflexion</u>

(M-I)



bcoz in the Left Nbd of $x=0$ it is Concave Downward
& " Right " " " Concave upward.

(M-II) $f(x)=x^3$, $f'(x)=3x^2$, $f''(x)=6x$, $f'''(x)=6$

Put $f''(x)=0$

$6x=0$

i.e $x=0$

Now $f'''(0)=(6)_{x=0}=6\neq 0$

$\Rightarrow x=0$ is Point of Inflexion.

Note:- Also find the T. Point of $f(x)=x^3$, Sol:- $f'(x)=0 \Rightarrow 3x^2=0 \Rightarrow x=0$

Q: Find the Inflexion Points of $y=f(x)=(x^2-4)^2$, $D_f=(-\infty,\infty)$

Sol: $f(x)=(x^2-4)^2$

$f'(x)=2(x^2-4)(2x)$

$=4x^3-16x$

$=4x(x+2)(x-2)$

ie T Points are $x=-2,0,2$

$f''(x)=12x^2-16$

$f'''(x)=24x$

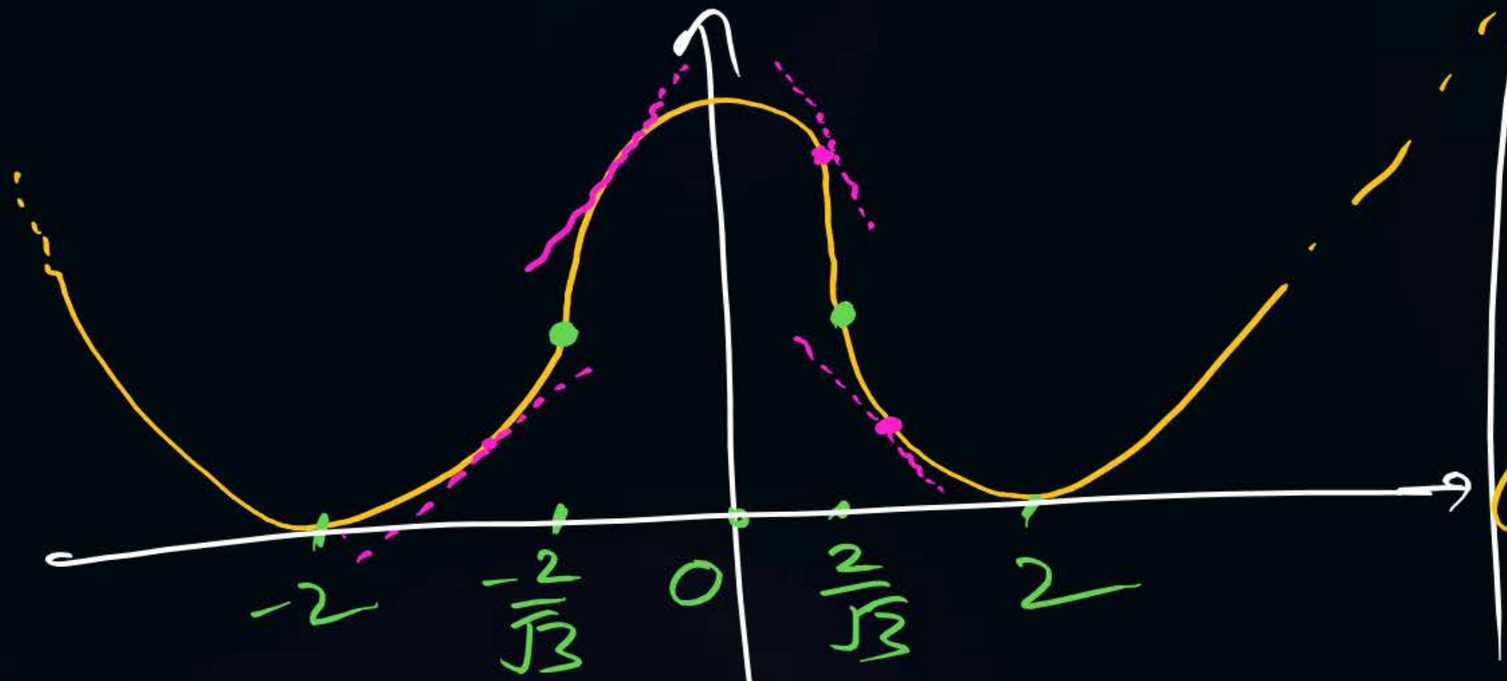
For Inflexion Points, Put $f''(x)=0$

$12x^2-16=0 \Rightarrow x=\pm\frac{2}{\sqrt{3}}$

Now $f'''\left(\frac{2}{\sqrt{3}}\right)=(24x)_{x=\frac{2}{\sqrt{3}}}\neq 0$

$f'''\left(\frac{-2}{\sqrt{3}}\right)=(24x)_{x=\frac{-2}{\sqrt{3}}}\neq 0$

ie Both $x=\frac{2}{\sqrt{3}}$ & $x=\frac{-2}{\sqrt{3}}$ are Points of Inflexion



Qs Find the Interval in which $f(x)=(x^2-4)^2$ is Concave upward & Concave Downward ?

(M-I) $f(x)$ S Inc in $(-2,0)\cup(2,\infty)$

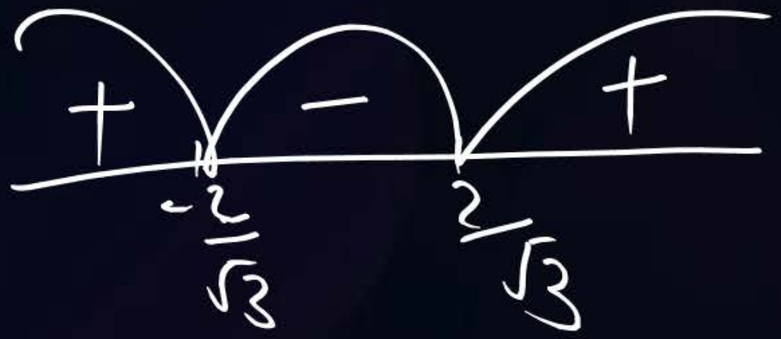
$f(x)$ S Dec in $(-\infty,-2)\cup(0,2)$

$f(x)$ is Concave up. in $\left(-\infty,-\frac{2}{\sqrt{3}}\right)\cup\left(\frac{2}{\sqrt{3}},\infty\right)$

(M-II) $f(x)=(x^2-4)^2$, $f'(x)=\boxed{4x(x+2)(x-2)}$

$f''(x)=12x^2-16=12\left(x^2-\frac{4}{3}\right)=\boxed{12\left(x-\frac{2}{\sqrt{3}}\right)\left(x+\frac{2}{\sqrt{3}}\right)}$

$f(x)$ is Concave Down in $\left(-\frac{2}{\sqrt{3}},\frac{2}{\sqrt{3}}\right)$



Sign of $f''(x)$

Q: Find the interval in which $f(x)=x^4-24x^2+11$ is S.Inc, S.Dec, Concave upward & Concave Downward ?

Sol: $f(x)=x^4-24x^2+11$, $D_f=(-\infty,\infty)$

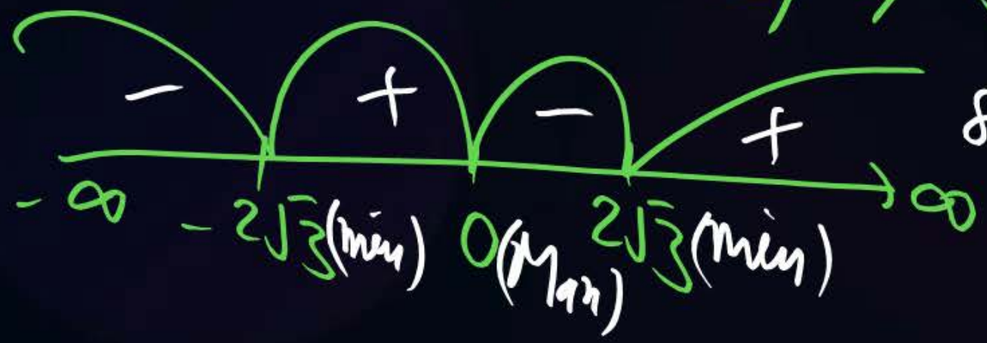
$f'(x)=4x^3-48x=4x(x^2-12)$

$f''(x)=12x^2-48=12(x^2-4)$

$f'(x)=4x(x-2\sqrt{3})(x+2\sqrt{3})$

So T. Points are, Put $f'(x)=0$

$\Rightarrow x=-2\sqrt{3},\ 0,\ 2\sqrt{3}$



So $f(x)$ is S.D in $(-\infty,-2\sqrt{3})\cup(0,2\sqrt{3})$

$f(x)$ is S.Inc in $(-2\sqrt{3},0)\cup(2\sqrt{3},\infty)$

Now for Inflexion points, Put $f''(x)=0$

$12(x^2-4)=0\Rightarrow x=-2,2$



$f(x)$ is Concave upward in $(-\infty,-2)\cup(2,\infty)$

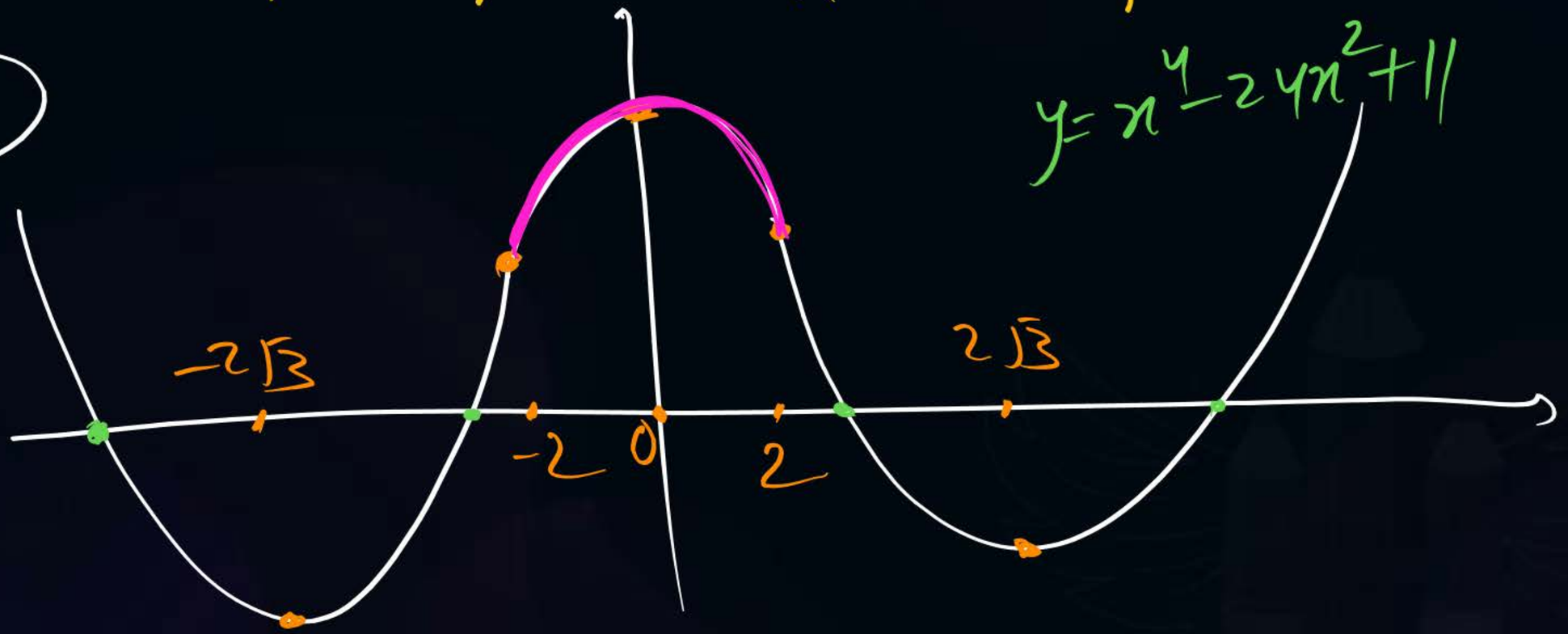
$f(x)$ is Concave Downward in $(-2,2)$

In previous Ex, T. Points are $x = -2\sqrt{3}, 0, 2\sqrt{3}$

& Saddle points are $x = -2$ & 2

M-II

$y = x^4 - 24x^2 + 11$



Q. the Number of Inflexion Points of $f(x) = x + 12x^4$ is/are?

(PYQ)

$f(x) = x + 12x^4$

$f'(x) = 1 + 48x^3$

$f''(x) = 144x^2$

$f'''(x) = 288x$

Put $f''(x) = 0$

$144x^2 = 0$

$x = 0$

Now $f'''(0) = (288x)_{x=0} = 0$

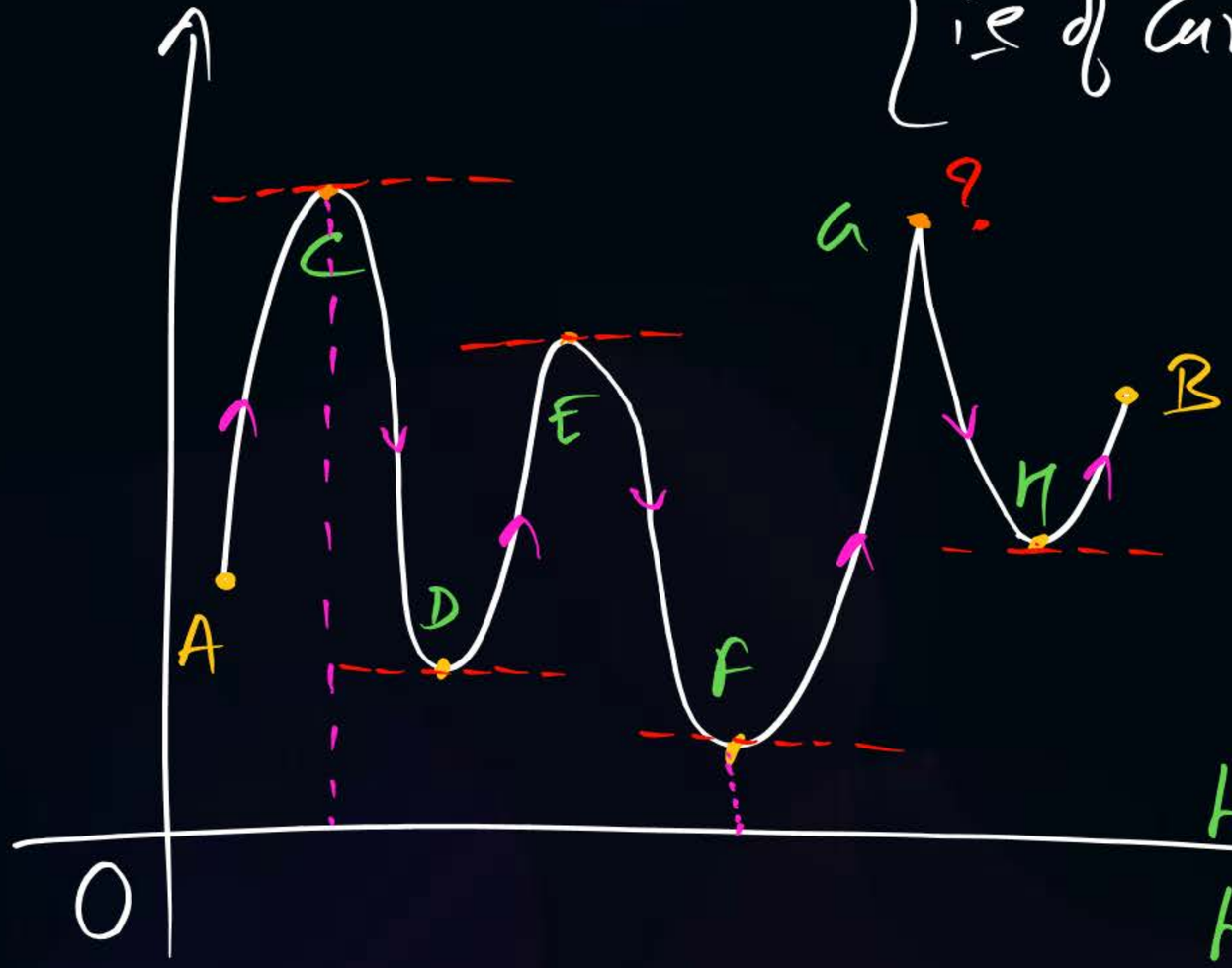


i.e $x = 0$ is not a Saddle point.

So No. of Inflexion Points = ZERO

MAXIMA-MINIMA of Func^n of Single Variable

[i.e of Curve $y=f(x)$] → 12th Class.



Point $\simeq x$

Value $\simeq y$

Local Max Points → C, E, G, B

Local Max Values → f(C), f(E), f(G), f(B)

Absolute Maximum / Global → f(C)

Local min Points → A, D, F, H

Local min Values → f(A), f(D), f(F), f(H)

Absolute minima / Global → f(F)

① Necessary Condition for Maxima-Minima → $f'(x) = 0$ or Not Defined — (1)

② Stationary Points: eg C, D, E, f, H

those points where tangent is Horizontal are called Stationary Points.

ie at Stationary Points, $f'(x) = 0$

③ Turning Points / critical Points → eg C, D, E, f, H, G

those points where tangent is either Horizontal or Not defined as Called T. Points. ie at T. Points, $f'(x) = 0$ or undefined (T)

→ All the Stationary Points are Turning Points But Converse is not Necessarily True

→ By solving eqn (1) we can find Turning Points.

T-Points.

④ Extreme Points / Optimal Points → eg A, C, D, E, F, G, H, B

those points where we get either Maxima or Minima are called E-Points

→ Corner Points are also Extreme Points (True)

But take Care; Corner Points are not Critical Points.

⑤ Points of Inflexion / Saddle points →

those points where Curve Changes it's Concavity are called Saddle points

⑥ Maxima & Minima occurs alternately (True)

⑦ Sufficient Cond^{n} for Maxima-Minima → f(x) must satisfy either

1st Derivative test or 2nd Derivative test.

(8) 1st Derivative test :- Let $x = c$ is the Turning Point.

(it is obtained by solving equ" (1)) i.e. $f'(c) = 0$ or undefined

Now we will check the sign of $f'(c)$ as follows; M. Imp.



(Maxima)



(minima)

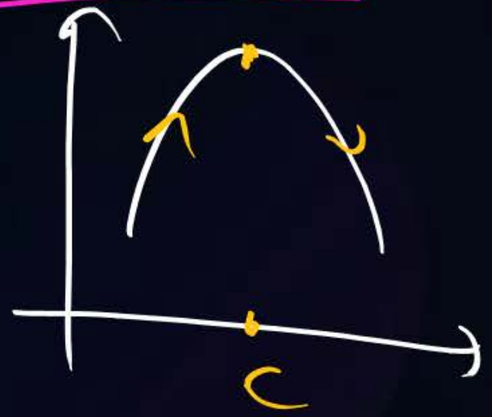


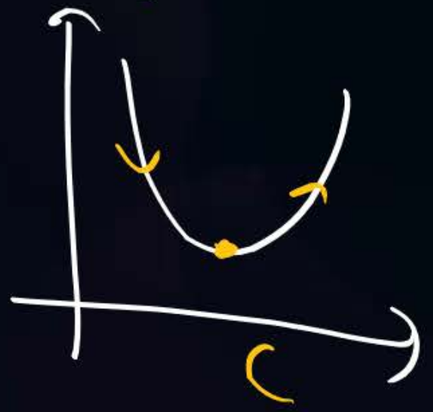
(NNNM)



(NMNM)

Explanation :-









(9) 2nd Derivative test → Let $x=c$ is Critical Point then

$f'(c)=0$

If $f^n(c)=0$ & $f^{n+1}(c)\neq 0$;
n = even then
$x=c$ is Point of Inflexion

$f''(c)>0$

($x=c$ is Point of Minima)

$f(x)$ is minimum at $x=c$

$f''(c)=0$

$f'''(c)\neq 0$

($x=c$ is Saddle point)

$f''(c)<0$

($x=c$ is Point of Maxima)

$f(x)$ is Maximum at $x=c$

(10) If we want to check the Nature of Corner points, then we have (No) Shortcut Method ☹

it's Nature can be Calculated by using following theorem only

"Maxima & Minima occurs Alternately"

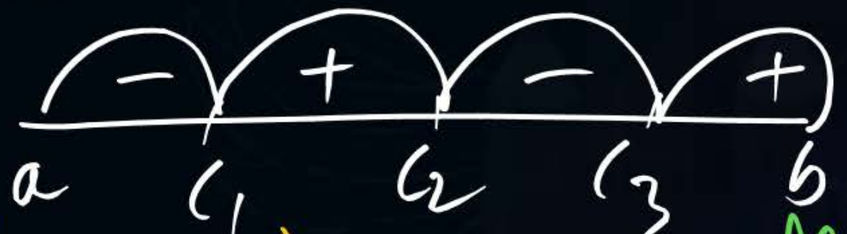
Eg Let $f(x)$ is defined in $[a,b]$ & it's T.Points are $x=c_1,c_2,c_3$

Case I:



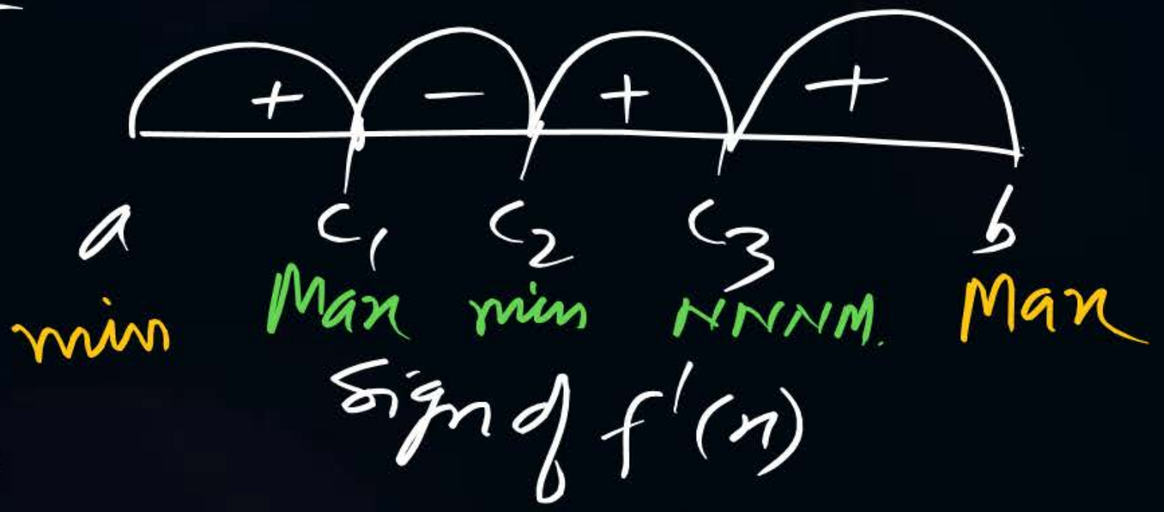
Sign of $f'(x)$

Case II:

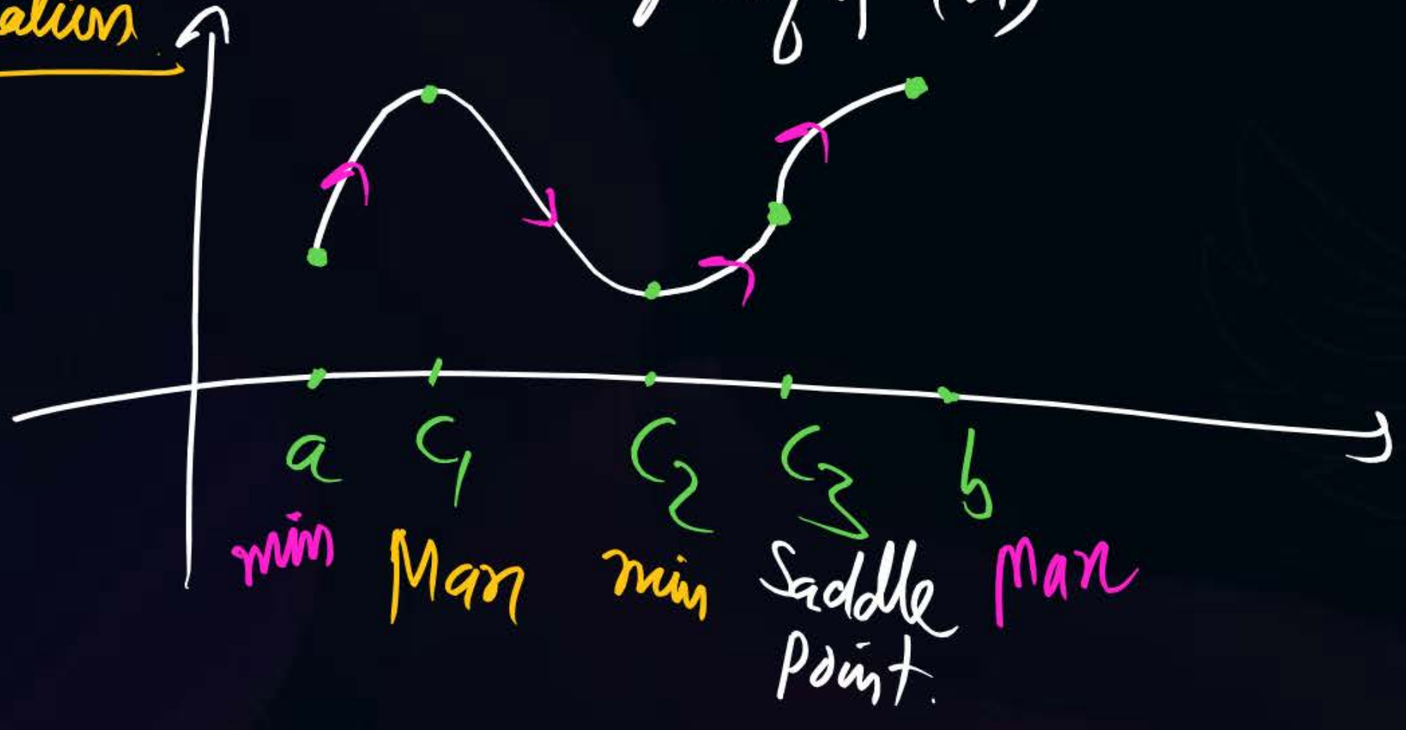


Sign of $f'(x)$

Case II :

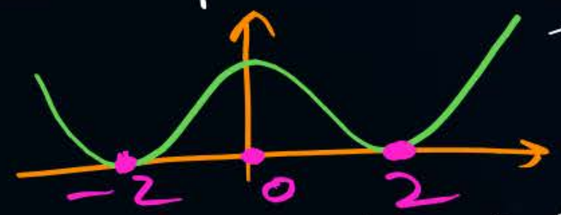


Explanation :



(11) An nth degree polynomial Bends at Most $(n-1)$ times
So it has at Most $(n-1)$ extremas (Maxima & Minima both)

eg $y=(x^2-4)^2$ (it is polynomial of degree 4)



$\Rightarrow$ It has three extremas (one Local Maxima & Two L. Minima)

Note: A cubic polynomial with Real Coefficients has at Most 2 extremas & 3 Zero Crossings.
(Maxima & minima both) (Roots).

THANK - YOU

Tel: drbunnet Sir pw